

Turbo Product Code-Aided Spatiotemporal 2-D Coded MIMO for URLLC

Yu Tian, Jian Zheng, Zongyao Li, Tiancan Xia, Yifei Shen, Wenyue Zhou, Xiaosi Tan, Jiamin Li, Yongming Huang, Chuan Zhang, and Xiaohu You

ABSTRACT

The ultra-reliable low-latency communication (URLLC) scenario beyond the fifth-generation mobile networks requires latency to shift from ms-level to μ s-level while maintaining low packet error rates. Short block-length codes can reduce latency but suffer from performance degradation. Asymptotic analyses indicate that increasing spatial degrees of freedom in multiple-input multiple-output (MIMO) systems help to achieve target error rates with a fixed block length and signal-to-noise ratio. To approach the upper bound of the average maximal achievable information rate in massive MIMO systems, spatiotemporal 2-D channel coding has been proposed. However, a considerable gap remains between the performance of the existing coding scheme and the theoretical capacity bound. To bridge this gap, this article introduces turbo product codes aided spatiotemporal 2-D coded MIMO systems (TPCAS-MIMO). Multi-level design on the baseband MIMO transceiver, involving encoding, interleaving, modulation, detection, and decoding schemes, is explored to boost the performance of TPCAS-MIMO with low latency for URLLC. In comparison to prior arts, this work achieves a 5.5 dB SNR gain at a frame error rate of 10^{-5} using $(64, 42)^2$ polar turbo product codes in a 64×128 MIMO system, narrowing to a mere 2.5 dB gap from the theoretical capacity bound.

INTRODUCTION

The design of the physical layer for ultra-reliable low-latency communication (URLLC) faces substantial challenges in satisfying two inherently conflicting demands: ultra-high reliability (10^{-5} to 10^{-9}) and low end-to-end latency of less than 0.1 ms [1]. The fifth-generation (5G) mobile networks struggle to meet certain industrial and medical applications with a sub-0.1 ms latency requirement [2]. Short packets can be employed to reduce latency, which in turn causes a severe loss in coding gain. One case of this trade-off is low-density parity-check (LDPC) codes selected for the enhanced mobile broadband (eMBB) scenario of 5G. When extending eMBB channel coding to URLLC, [3] reveals an error floor in LDPC codes for short block scenarios in eMBB. For the upcoming 6G, one focus is on achieving URLLC with latency in the microsecond

(ms) range, which necessitates the development of channel coding schemes capable of achieving high reliability even with short blocklengths. It has been indicated that further research is needed on channel coding for URLLC with short information blocks [3]. Classical Shannon information theory exhibits a performance trade-off between information rate, latency, and reliability, which hinders scenarios with short blocks [4]. In finite blocklength coded multiple-input multiple-output (MIMO) systems, a similar rate-latency-reliability trade-off is illustrated in [5], referred to as the phenomenon of capacity collapse. Fortunately, [5] indicates that target error rates for a fixed block length and signal-to-noise ratio (SNR) can be achieved by increasing spatial degrees of freedom (DoFs) in MIMO systems. Therefore, massive MIMO systems, equipped with numerous antennas to enhance DoFs, promise to surpass the rate-latency-reliability trade-off and enable URLLC. To leverage the DoFs, spatiotemporal 2-D channel coding [6] has been proposed and can be a solution to approach the upper bound of the average maximal achievable information rate of massive MIMO systems. In spatiotemporal coding, temporal coding corresponds to 1-D channel coding, while spatial coding spans multiple bitstreams. The receiver employs joint decoding in both time and space domains for enhanced system performance, with parallel decoding of each codeword ensuring low latency. Despite the advantages of low latency, the traditional spatiotemporal 2-D coded MIMO systems in [6] fall short of bridging the gap with the theoretical capacity bound. There is a compelling need for a novel enabling framework to enhance the error-correction performance of spatiotemporal 2-D coded MIMO systems.

In spatiotemporal 2-D coded MIMO systems, when each row and column of the codeword trellis corresponds individually to a codeword with the same properties, the encoding process is equivalent to that of turbo product codes (TPCs). TPCs provide near-Shannon-limit performance [7] and are included in various communication standards such as the IEEE 802.16. Recently, polar-based TPCs (polar-TPCs) [8] have received considerable attention for 400G-ZR specification in optical systems due to superior performance compared to conventional TPCs based on Bose-Chaudhuri-Hoc-

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quenghem (BCH) codes. The performance degradation caused by the short lengths of constituent polar codes can be mitigated by turbo iterations. This allows the system to approach the near-optimal performance of a long, uncoupled polar code. The soft-output decoding of TPCs is promising to enhance the performance of spatiotemporal 2-D channel coding. However, the straightforward application of TPCs in spatiotemporal 2-D coded MIMO systems faces two key challenges. One challenge is the constraint in maximum iteration number during TPC decoding, which requires careful selection of appropriate component codes of TPCs for optimal performance with low latency to enable URLLC. The other challenge is the burst noise occurring in both time and space domains of high-order modulated MIMO systems, which can be mitigated by carefully designing an interleaver. The latency introduced by a traditional TPC interleaver, proportional to block lengths, contradicts the latency requirements of URLLC. To address the above-mentioned challenges, we introduce TPCs-aided spatiotemporal 2-D coded MIMO systems (TPCAS-MIMO).

The main contributions of this work are listed as follows.

- We select ultra-short polar codes as component codes of TPCs in the proposed TPCAS-MIMO and reveal that polar codes surpass the commonly used BCH codes in terms of performance and convergence at identical code lengths and comparable code rates.
- We propose time-domain interleaving, which is the delay related to the code length in the time domain. These strategies achieve a remarkable 98.7 percent delay reduction compared to uniform interleaving while maintaining comparable performance.
- We illustrate that the proposed TPCAS-MIMO provides a 5.5 dB SNR gain at a frame error rate (FER) of 10^{-5} compared to the traditional spatiotemporal 2-D MIMO systems [6], with only a 2.5 dB gap to the theoretical bound. Compared to 1-D coded MIMO systems with long block lengths, TPCAS-MIMO shows a 0.83 dB performance gain with a significant 79.6 percent reduction in normalized decoding time steps.

The remainder of this article is organized as follows. The next section reviews the existing layer mapping schemes and spatiotemporal 2-D coded MIMO system. We then introduce the proposed TPCAS-MIMO framework and describe its components. Following that, we present numerical results. The final section concludes the entire article and highlights future research challenges.

RELATED WORKS

In this section, we overview the typical layer mapping schemes, specifically the folded and parallel layer mapping. We also discuss the spatiotemporal 2-D channel coding, which ingeniously integrates these two mapping schemes.

LAYER MAPPING

Layer mapping in 5G involves the allocation of data streams or information bits to physical resources, such as antennas and frequency subcarriers, during the transmission process. Effective layer mapping enhances the efficiency of the sys-

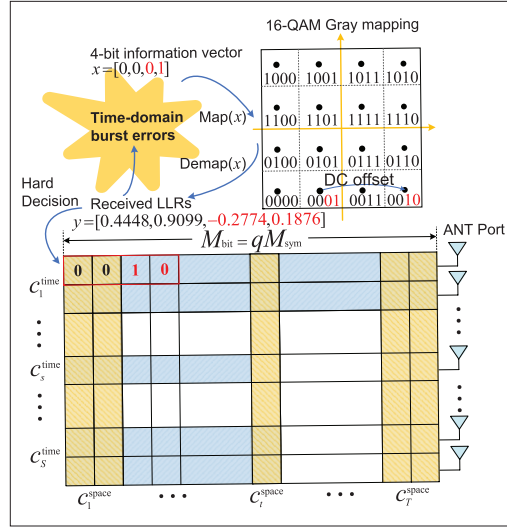


FIGURE 1. The codeword trellis in spatiotemporal 2-D channel coding and 16-QAM constellation symbols labeled with Gray code.

tem by maximizing the use of available resources and adapting to varying network conditions, ensuring high data rates and reliability in URLLC. Folded layer mapping allocates layers to a single physical channel, which can optimize resource usage by combining multiple streams into fewer physical resources. On the other hand, parallel layer mapping assigns each data layer to separate physical channels, allowing for the simultaneous transmission of multiple streams, albeit potentially requiring more resources.

SPATIOTEMPORAL 2-D CHANNEL CODING

Spatiotemporal 2-D channel coding in MIMO systems introduces inter-stream (space domain) coding over intra-stream (time-domain) coding [6]. The nature of spatiotemporal 2-D coding, such as high parallelism and large minimum Hamming distance, makes it promising to meet the requirements of URLLC. The layer mapping scheme, shown in Fig. 1, combines folded and parallel mapping schemes. In each layer, M_{sym} complex-valued symbols are mapped from M_{bit} coded bits. The encoding is categorized into time-space (T-S) mode and space-time (S-T) mode. In the T-S mode, each bitstream $\mathbf{b}_i$ ($i = 1, 2, \dots, S$) in the time domain is first encoded to the codeword $\mathbf{c}_i^{\text{time}}$ with code length N_i and code rate R_i . Then, $\mathbf{c}_i^{\text{time}}$ is mapped to N_i/M_{bit} layers with the assistance of rate matching as demonstrated in Fig. 1, where M_{bit} is the time-domain length of spatiotemporal 2-D codeword. Similarly, each codeword $\mathbf{c}_j^{\text{space}}$ ($j = 1, 2, \dots, T$) is encoded in the space domain. For each codeword $\mathbf{c}_j^{\text{space}}$, the length and the rate are N_j and R_j , respectively. Conversely, the S-T mode performs the space-domain coding first, then the time-domain coding.

THE PROPOSED FRAMEWORK FOR SPATIOTEMPORAL 2-D CHANNEL-CODED MIMO SYSTEMS

In this section, we improve the classical spatiotemporal 2-D channel-coded MIMO systems by introducing a TPCs-aided spatiotemporal 2-D coded MIMO system (TPCAS-MIMO). TPCAS-MIMO maps codewords in space and time domains

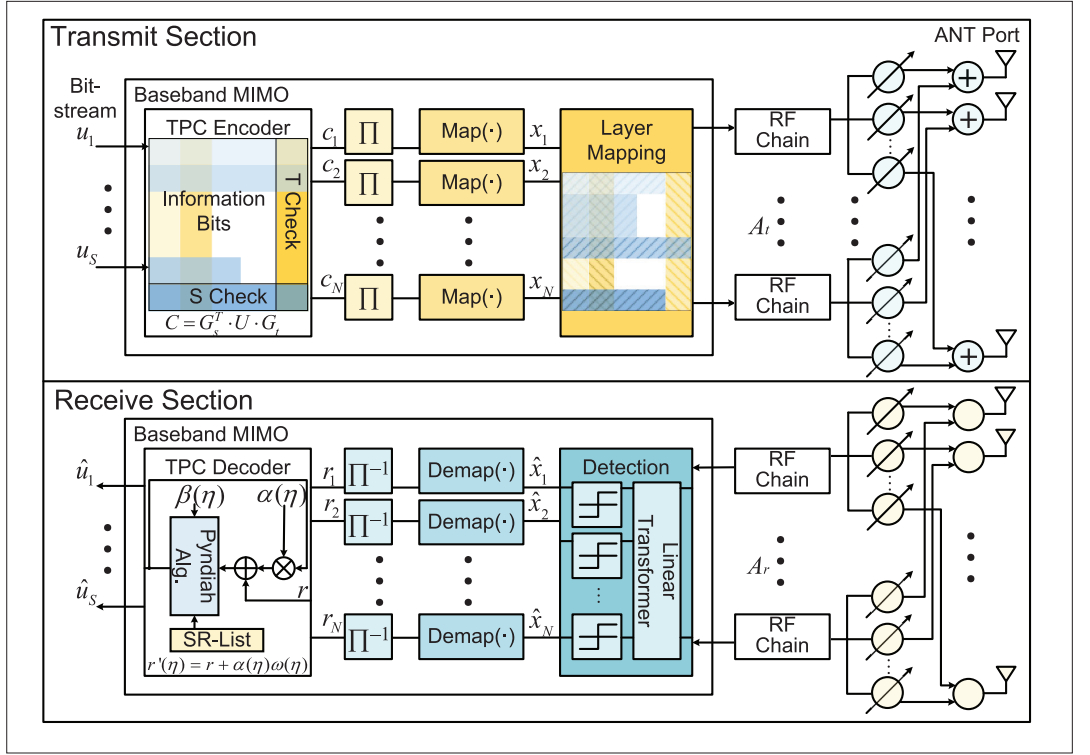


FIGURE 2. The block diagram of the multi-antenna baseband transmitting system with the TPCAS-MIMO framework.

through a TPC manner, where the component codes adopt short polar codes. Additionally, we propose a low-complexity interleaving scheme tailored to TPCAS-MIMO. Furthermore, concerning the decoder, we introduce a soft-input soft-output iterative decoding using sequence repetition list (SR-List) decoding to achieve improved performance and low latency. Figure 2 demonstrates the block diagram of the multi-antenna baseband transmitting and receiving system with the TPCAS-MIMO framework. We provide detailed explanations of the encoding, interleaving, modulation, detection, and decoding schemes of TPCAS-MIMO as follows.

ENCODING

In TPCAS-MIMO systems, each codeword in the same domain shares the same code type, rate, and length, as described for the construction of TPCs. TPCs are serially concatenated codes constructed by two or more component block codes [7]. Let us consider a 2-D TPC $\mathcal{C}$ constructed by linear block codes $\mathcal{C}_i$, $i = 1, 2$ with the parameters $(n_1 \times n_2, k_1 \times k_2, d_1 \times d_2)$, where n_i , k_i , and d_i denote the code length, information length, and minimum distance of $\mathcal{C}_i$, respectively. Conventional TPCs are typically constructed using algebraic codes such as BCH codes [7]. However, [9] has reported that TPCs constructed by polar codes exhibit superior error correction performance and convergence characteristics over those based on BCH codes with equal code length and rate. A polar code $\mathcal{P}(N = 2^m, K)$ is a binary linear block code generated by the polarization transformation matrix $\mathbf{G} = \mathbf{G}_2^{\otimes m}$, where $\otimes m$ denotes the m -times Kronecker product of the kernel matrix

$$\mathbf{G}_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

The codeword is given by $\mathbf{c} = \mathbf{u}\mathbf{G}$ with the information bits allocated to the K most reliable bit positions in $\mathbf{u}$ and the remaining $N - K$ bits set to zero in the positions listed in a frozen set $\mathcal{F}$. Successive cancellation (SC) decoding aided by cyclic redundancy check (CRC) is known for its effectiveness for short block lengths. With a large list size, CRC-aided SC decoding could achieve near-maximum likelihood (ML) decoding performance.

Given two component polar codes $\mathcal{C}_t$ and $\mathcal{C}_s$ with parameters $(N_t, K_t, \mathcal{F}_t)$ and $(N_s, K_s, \mathcal{F}_s)$ respectively, spatiotemporal 2-D coding allows to construct a code $\mathcal{C}$ of length $N = N_s \times N_t$ and dimension $K = K_s \times K_t$. Encoding starts from an $N_s \times N_t$ matrix $\mathbf{U}$, containing K information bits. In the T-S mode, each codeword $\mathbf{c}_t$ is initially encoded independently by appending $N_t - K_t$ time-domain (T) check codes. Then, the resulting $\mathbf{U}_t$ is encoded using $\mathcal{C}_s$ with $N_s - K_s$ space-domain (S) check codes. In the resultant $N_s \times N_t$ codeword matrix $\mathbf{C}$, rows and columns are codewords of $\mathcal{C}_t$ and $\mathcal{C}_s$, respectively. The encoding procedure can be mathematically expressed as

$$\mathbf{C} = \mathbf{G}_s^T \cdot \mathbf{U} \cdot \mathbf{G}_t, \quad (1)$$

where $\mathbf{G}_s$ and $\mathbf{G}_t$ are the generator matrices of codes $\mathcal{C}_s$ and $\mathcal{C}_t$, respectively. It is worth noting that when interpreting spatiotemporal 2-D channel coding as TPCs, reversing the encoding order does not alter the resulting codeword matrix. Therefore, in the encoding process of TPCAS-MIMO, the T-S and S-T modes are interchangeable.

INTERLEAVING AND MODULATION

We consider high-order quadrature amplitude modulation (QAM) due to its high spectral efficiency in low-noise channel conditions and wide applications in wireless communications systems.

In practice, q -bit Gray codes are mapped to constellation symbols. However, high-order modulated symbols are susceptible to direct current (DC) offset [10]. The consequential shifts induced by the DC offset have the potential to precipitate burst errors when they compel the signal to cross decision thresholds during the demodulation process, which is illustrated by the following example.

Consider that a symbol, formed by four information bits: {0, 0, 0, 1}, is transmitted, and the log-likelihood ratios (LLRs) vector of [0.4448, 0.9099, -0.2774, 0.1876] is received. As shown in Fig. 1, these localized errors pose a substantial challenge for error correction mechanisms. Interleaving mixes code symbols from different codewords, spreading error bursts across multiple codewords during reconstruction. The process of interleaving and deinterleaving awaits sufficient data for effective rearrangement, which entails an undesired latency even for short-length high-rate codes in spatiotemporal 2-D coded MIMO systems.

Inserting a uniform interleaver between two component codes of a TPC is the conventional solution to improve the convergence of iterative soft-decision decoding algorithms. Unfortunately, it falls short of meeting the latency requirements of URLLC. The column-interleavers can preserve the minimum distance of the product code by interleaving only one of the two-component codes [11]. The spatial codes associated with different subcarriers are independent in orthogonal frequency division multiplexing (OFDM) systems. In contrast to the heavily correlated errors in the time domain, the occurrence of burst errors in space-domain codes is notably infrequent, as illustrated in Fig. 1, which renders it feasible to obviate interleaving in the space domain and thus enable URLLC. In our analysis, we concentrate on the triangular interleaver, which has been proposed in 5G standards as the channel interleaver [12]. This interleaver has been considered necessary to improve the coding performance for high-order modulation. The encoded bits of length N are arranged into rows within an isosceles triangular structure of length T bits with

$$T = \left\lceil \frac{\sqrt{8N + 1} - 1}{2} \right\rceil.$$

The interleaved vector is obtained by reading bits from the structure in columns. The interleaver/deinterleaver delay is approximately

$$\frac{T(T+1)}{2}$$

bit times, and the memory requirement is $2T^2$ bits.

MASSIVE MIMO SYSTEM MODEL AND DETECTION

We consider a massive MIMO system equipped with A_t transmit antennas and A_r receive ANT ports. The modulated symbols $\mathbf{X}$ are transmitted over the massive MIMO channel, and the receiver signal is

$$\mathbf{Y} = \mathbf{H} \cdot \mathbf{X} + \mathbf{N}, \quad (2)$$

where $\mathbf{H}$ is the massive MIMO channel matrix and the vector $\mathbf{N}$ denotes the additive white Gaussian noise (AWGN).

The receiver recovers the transmitted signal vector $\mathbf{X}$ from $\mathbf{Y}$ based on detection techniques. In this article, we assume perfect channel state

information at the receiver end. The complexity of the optimal detection algorithm grows exponentially with antennas, making it unsuitable for massive MIMO. Linear detection algorithms such as minimum mean square error (MMSE) are widely adopted due to their low complexity, aligning with the latency requirements of URLLC. However, to fulfill ultra-high reliability demands, there is room for some high-performance nonlinear detection algorithms. Among them, belief-selective propagation (BsP) detection [13] and expectation propagation (EP) detection [14] strike a balance between complexity and performance. Depending on the specific scenario requirements, three detection configurations are provided for TPCAS-MIMO systems.

MMSE Detector: The MMSE detector obtains the estimation of transmitted symbols with minimized mean square error (MSE). The boundary judgment method is usually used for bit LLR calculation to ensure a low-complexity system.

Expectation Propagation (EP) Detector: The basic idea of EP is to approximate an intractable posterior probability $p(\mathbf{x})$ using a Gaussian distribution $q(\mathbf{x})$, thus minimizing the Kullback-Leibler divergence $D_{KL}(p(\mathbf{x}) || q(\mathbf{x}))$. It was shown in [14] that this approach amounts to the moment matching condition, wherein the first and second-order moments of $p(\mathbf{x})$ and $q(\mathbf{x})$ match each other.

Belief-Selective Propagation (BsP) Detector: The BsP detector [13] is a revised version of the belief propagation (BP) detector. By employing information with substantial a priori probabilities selectively when updating messages, BsP encounters an improved performance with an even lower complexity than the conventional BP detector.

ITERATIVE DECODING

After MIMO detection and deinterleaving (the inverse process of interleaving), the obtained bit LLRs matrix $\mathbf{R}$ (size $N_s \times N_t$) serves as the initial input for the decoder. TPCAS-MIMO performs decoding alternately in the time and space domain using Pyndiah's algorithm [7] with SR-List [15]. As shown in Fig. 2, the SR-List decoders first generate candidate codewords for each row or column of $\mathbf{R}$ (denoted as $\mathbf{r}$) to calculate the extrinsic information $\omega(\eta)$ using the parameter β in Pyndiah's approximation algorithm. In the next iteration, the input LLRs $\mathbf{r}'(\eta)$ for the SR-List decoders are updated with the newly calculated $\omega(\eta)$ and the parameter α . The iteration process is terminated upon reaching the maximum number of iterations $\eta_{\max}$. The decoding latency of SR-List can be evaluated by the count of decoding time steps $\mathcal{T}(\mathcal{F})$. In each half-iteration, parallelly performing time (space) domain decoding requires $\mathcal{T}(\mathcal{F}_t)$ ($\mathcal{T}(\mathcal{F}_s)$) time steps, while computing the extrinsic information requires an additional 2 time steps. Thus, the decoding latency of TPCAS-MIMO is

$$D_{\text{iter}} = \eta_{\max} \cdot (\mathcal{T}(\mathcal{F}_s) + \mathcal{T}(\mathcal{F}_t) + 4), \quad (3)$$

which is dominated by the maximum number of iterations $\eta_{\max}$ as a trade-off for performance improvement.

In high-order modulated MIMO systems, as discussed previously, modulation introduces time-domain burst noise. In the spatial domain, the burst noise is induced by inter-antenna inter-

Inserting a uniform interleaver between two component codes of a TPC is the conventional solution to improve the convergence of iterative soft-decision decoding algorithms. Unfortunately, it falls short of meeting the latency requirements of URLLC.

EXPERIMENTAL RESULTS

In this section, we evaluate the performance and latency of TPCAS-MIMO through Monte Carlo simulations and analysis. The simulations are conducted under 64×128 MIMO systems. CRC-11 with polynomial $0 \times E21$ is aided by 1-D polar codes, and these codes are decoded by the SR-List algorithm. We do not attach CRC to the component polar codes since their codelengths are ultra-short. Unless specified otherwise, default scenarios utilize 16-QAM modulation and MMSE detection with separate detection and decoding (SDD). The values of α and β for the first ten half iterations are optimized experimentally based on the optimization strategy in [7], namely

$$\alpha = [0, 0.30, 0.30, 0.45, 0.50, 0.60, 0.70, 0.75, 0.80, 0.85], \\ \beta = [0.2, 0.40, 0.60, 0.80, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00].$$

A 2-D code characterized by parameters (N_t, K_t) and (N_s, K_s) is denoted as $(N_t, K_t) \times (N_s, K_s)$. With $N_t = N_s$ and $K_t = K_s$, the 2-D code is briefly denoted as $(N_t, K_t)^2$. When the number of transmit antennas is not a power of two, we adopt the code-block segmentation method outlined in [6], where the antenna array is divided into multiple codewords. For generality and clarity, we focus on the scenario where the number of transmit antennas corresponds to the power-of-two code length in the space domain to simplify the analysis.

COMPONENT CODES COMPARISON FOR TPCAS-MIMO

First, we analyze the advantages of using polar codes as component codes in TPCAS-MIMO for URLLC scenarios. Figure 3 depicts the FER performance comparison of TPCAS-MIMO with polar codes and extended BCH (eBCH) codes serving as the component codes under the same number of candidate codewords. The decoding algorithm for eBCH-TPCs is the widely used Chase-Pyndiah algorithm [7], with the number of least reliable bits set to 6, which generates $2^6 = 64$ candidate codewords. The L candidate codewords generated by an SCL algorithm with a list size of L are distinct and valid. However, in contrast, Chase-Pyndiah generates candidate codewords inefficiently, including redundant codewords and invalid codewords. In our simulations, the unique and valid candidate number is roughly 8, which is the same as the list size of SCL. Results demonstrate that polar-TPCs obtain superior performance than eBCH-TPCs under the same iteration number. Polar-TPCs require fewer iteration numbers to converge than eBCH-TPCs. The above results illustrate that polar-TPCs can gain more reliability with lower latency compared to eBCH-TPCs.

BURST NOISE AND INTERLEAVING

Figure 4 illustrates the performance of $(64, 42)^2$ polar-TPC with Reed-Muller (RM) construction under various interleaving schemes within the proposed TPCAS-MIMO. For a single-input single-output (SISO) system with AWGN channels, the FER curves overlap for both T-S and S-T modes when using uniform interleaving with the maximum number of iterations $\eta_{\max} = 2$. In MIMO systems, nearly identical performance is observed for the two modes with uniform interleaving. Without interleaving, the S-T mode outperforms the T-S mode in both SISO and MIMO systems.

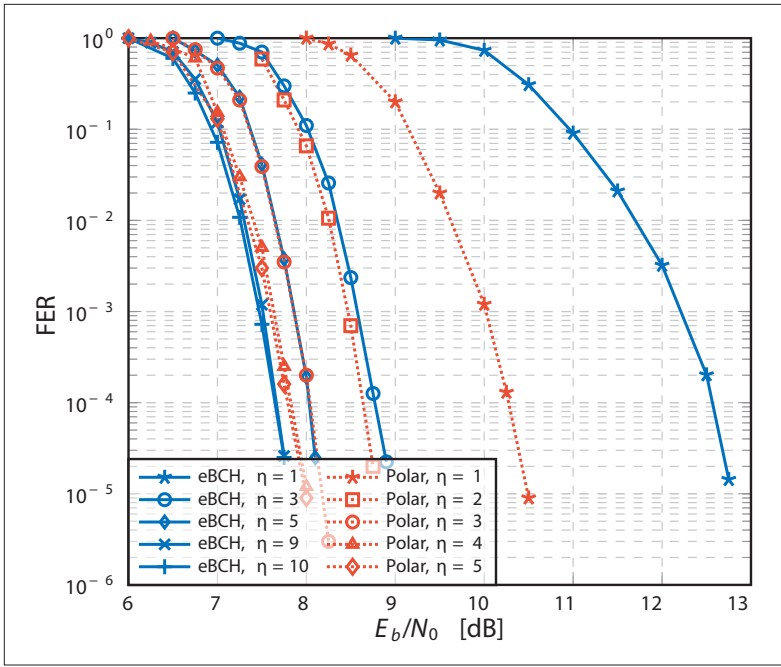


FIGURE 3. FER performance comparison of TPCAS-MIMO with a $(64, 42)$ Reed-Muller (RM)-construction polar code and a $(64, 45)$ eBCH code serving as the component codes under the same list size.

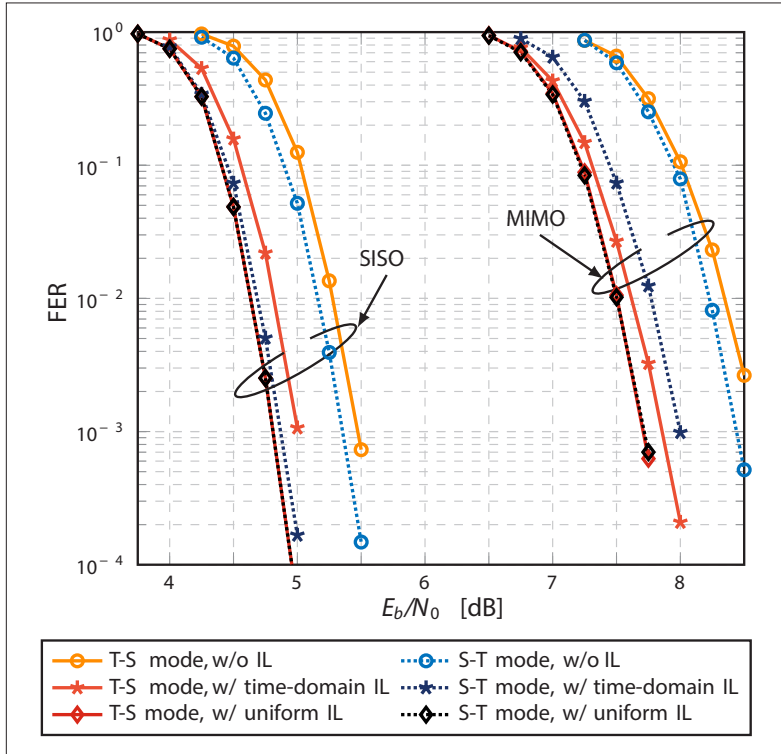


FIGURE 4. FER performance for 16-QAM SISO (left) and MIMO (right) systems with (w/) uniform interleaving (IL), time-domain IL, and without (w/o) IL.

ference. The turbo decoding mechanism of TPC offers a method for burst error propagation. To mitigate the impact of burst noise, decoding should commence from the dimension less affected by burst noise. Thus, despite the interchangeability of encoding order in TPCAS-MIMO, we differentiate the decoding methods based on the dimensional order: decoding starting with the spatial dimension is termed S-T mode, whereas the reverse is termed T-S mode.

This phenomenon further verifies that the incidence of burst noise in the space domain is significantly less frequent than in the time domain. With time-domain interleaving, the T-S mode outperforms the S-T mode in the MIMO system, contrary to the SISO system. This inconsistency can be attributed to the fact that while dispersed in the time domain, correlated errors persist in the space domain due to the properties of the Rayleigh fading channel. It should be noted that the time-domain interleaving with {T-S} mode approaches the uniform interleaving in the proposed TPCAS-MIMO. Compared to uniform interleaving, the proposed time-domain interleaving can save 6.48 percent storage and 98.65 percent delay with acceptable performance penalization.

PERFORMANCE AND LATENCY COMPARISON

Figure 5 demonstrates the comparison of the decoding performance of the TPCAS-MIMO versus 1-D and classical spatiotemporal 2-D coded MIMO systems [6]. The 5G LDPC and LTE Turbo codes, supported by 3GPP, share the same code length and rate as TPCAS. The 5G LDPC codes are CRC-16 aided (polynomial 0×1021) and decoded using the belief propagation algorithm with offset Min-Sum approximation (OMS) [3], where the offset is 0.25. The LTE Turbo codes are CRC-24B aided (polynomial $0 \times 864 \text{CFB}$) and decoded with the Max-Log-MAP algorithm [3]. Our evaluation reveals that TPCAS-MIMO demonstrates superior error correction performance in high SNR regions compared to all other simulated candidates. Specifically, at an FER of 10^{-5} , TPCAS-MIMO exhibits a gap of merely 2.8 dB from the normal approximate bound (theoretical bound) derived in [5]. Moreover, substituting MMSE with a more robust detection algorithm can notably improve error correction performance. When coupled with an EP or BsP detector using five iterations, the system achieves approximately a 0.70 dB performance gain at FER = 10^{-5} compared to the MMSE-based system.

Furthermore, the comparative analysis of normalized decoding time steps is depicted in Fig. 6. In this context, normalized decoding time steps refer to decoding time steps normalized by the number of information bits, thus serving as a quantifier of decoding relative latency. The LTE Turbo codes employ a radix-4 parallel MAP (PMAP) architecture [3] with a parallelism of 64, while the LDPC codes utilize an unrolled, fully parallel architecture [3]. Figure 6 reaffirms the optimal efficacy of setting $\eta_{\max} = 3$ for TPCAS-MIMO, striking a harmonious equilibrium between decoding performance and latency. Compared to the top-performing simulated 1-D codes, TPCAS-MIMO with $\eta_{\max} = 3$ achieves an impressive reduction of up to 79.6 percent in normalized decoding time steps. Despite its superior efficiency in normalized decoding time steps, the spatiotemporal 2-D coded MIMO system in [6] faces a significant 5 dB gap in error correction performance relative to TPCAS-MIMO. Hence, the proposed TPCAS-MIMO strikes a favorable position in the latency-reliability trade-off.

CONCLUSION AND FUTURE CHALLENGES

In this article, we highlight the nature of the spatiotemporal 2-D coding scheme as an application of TPCs and their variants in massive MIMO sys-

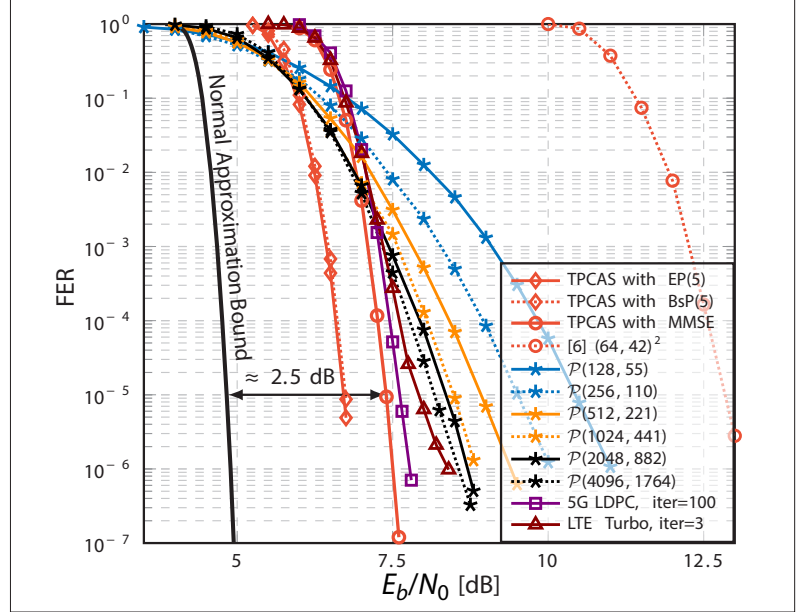


FIGURE 5. FER performance comparison for 16-QAM 64×128 MIMO systems with various channel coding schemes. Note: the list size is $L=8$ for list decoding based algorithms and CRC is only aided for 1-D decoding.

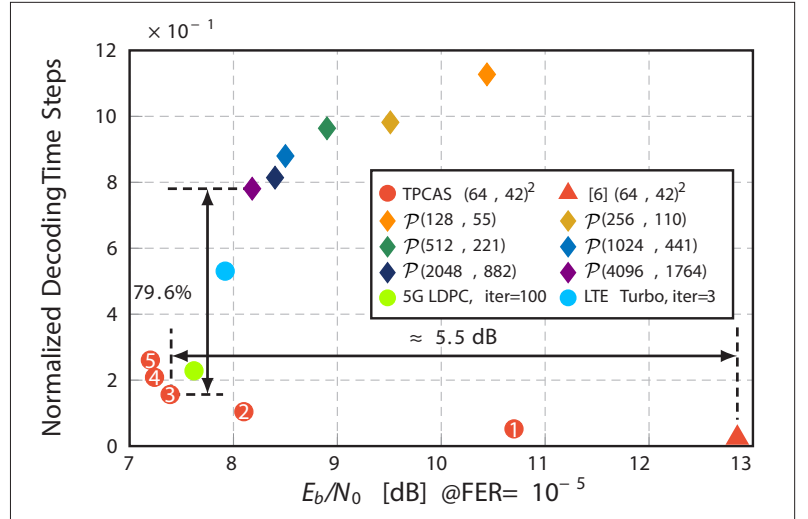


FIGURE 6. Decoding time steps comparison for various channel coding schemes @FER= 10^{-5} . Note: normalized decoding time steps are decoding time steps normalized by the number of information bits. The numbers on the markers of TPCAS-MIMO represent $\eta_{\max}$.

tems. According to this interpretation, Pyndiah's approximation for TPCs is incorporated into spatiotemporal 2-D coded MIMO systems to boost decoding performance. Moreover, we explore the baseband transceiver design on encoding, interleaving, modulation, detection, and decoding schemes. The proposed TPCAS-MIMO can be further tailored to meet performance and latency requirements for different application scenarios. Numerical results demonstrate that TPCAS-MIMO enables low latency while granting error correction performance for URLLC.

The future research will concentrate on the following points:

- *Performance Enhancement:* TPCAS-MIMO still remains 2.8 dB from the normal approximate bound, necessitating further research to develop efficient coding schemes.
- *Hardware Implementation:* Simulation results

With time-domain interleaving, the T-S mode outperforms the S-T mode in the MIMO system, contrary to the SISO system. This inconsistency can be attributed to the fact that while dispersed in the time domain, correlated errors persist in the space domain due to the properties of the Rayleigh fading channel.

indicate that TPCAS-MIMO reduces normalized decoding time by 79.6 percent. However, ultra-low processing delay relies critically on the hardware architecture of an appropriate receiver for the 2-D coded MIMO system, underscoring the necessity of efficient hardware implementation.

- *Joint Optimization Through Toolchains:* To achieve agile design, joint optimization, and efficient version iteration of the aforementioned theoretical optimizations and hardware implementations, we must utilize machine learning-assisted baseband circuit auto-generation toolchains.

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